OCSA-FN: A Fusion Network With Orthogonality-Constrained Spatial Attention for Hyperspectral and Land Surface Temperature Data Classification

Enyu Zhao, *Member, IEEE*, Yongfang Su, Nianxin Qu, *Student Member, IEEE*, Yulei Wang, *Member, IEEE*, and Yongguang Zhao

***Abstract*—With the continuous advancement of remote sensing (RS) technology, collaborative image land cover classification using multisource RS data has gradually emerged as a prominent research focus. However, one of the core challenges during the feature fusion process of multisource RS data is to achieve efficient feature interaction while minimizing redundancy and enhancing the representational capacity of features. To address this issue, this article proposes a fusion network with orthogonality-constrained spatial attention (OCSA-FN) for hyperspectral image (HSI) and land surface temperature (LST) data classification. First, OCSA-FN employs a dual-branch convolutional neural network (DB-CNN) module, where one branch utilizes a cross spatial-spectral CNN (CSS-CNN) to extract spectral features from HSI, while the other branch incorporates a learnable Sobel CNN (LS-CNN) to adaptively extract temperature features from single-channel LST data. Next, OCSA-FN introduces a dual-pooling residual channel attention (DRCA) module that leverages pooling-based interaction and residual connect to perform channelwise weighting on deep features. Subsequently, OCSA-FN presents an adaptive orthogonal feature fusion (AOFF) module designed to construct two sets of mutually orthogonal spatial basis vectors by imposing orthogonal constraints, and this effectively reduces the feature redundancy. Meanwhile, an adaptive spatial attention mechanism dynamically adjusts the fusion weights of the features between HSI and LST, facilitating efficient complementary fusion of multisource features. Finally, the weighted fused features are utilized for the classification. Extensive experimental results demonstrate that OCSA-FN outperforms state-of-the-art existing methods in terms of classification accuracy while reducing both the parameter count and computational complexity within the model.**

Received 5 July 2025; revised 22 September 2025 and 17 December 2025; accepted 25 January 2026. Date of publication 30 January 2026; date of current version 11 February 2026. This work was supported by the National Key Research and Development Program of China under Grant 2023YFB3905800 and Grant 2023YFB3905803. *(Corresponding authors: Yulei Wang; Yongguang Zhao.)*

Enyu Zhao, Yongfang Su, Nianxin Qu, and Yulei Wang are with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information Science and Technology College, Dalian Maritime University, Dalian, Liaoning 116026, China (e-mail: zhaoenyu@dlmu.edu.cn; 18009241387syf@dlmu.edu.cn; qvnx@dlmu.edu.cn; wangyulei@dlmu.edu.cn).

Yongguang Zhao is with the Key Laboratory of Quantitative Remote Sensing Information Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhaoyg@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3659827

***Index Terms*—Attention mechanism, hyperspectral image (HSI), land cover classification, land surface temperature (LST), orthogonal constraints.**

I. Introduction

REMOTE sensing (RS) images, acquired through satellites, aircraft, and various other platforms, serve as a vital data source for Earth observation [1]. Owing to the broad coverage, rich information dimensions, and strong timeliness, they have become a critical technology for surface monitoring and analysis. Land cover classification is one of the core research areas in RS image processing and has found widespread application in mineral exploration [2], environmental monitoring [3], urban planning [4], precision agriculture [5], and object detection [6], [7]. In earlier studies, land cover classification primarily relied on single data source [8]. Hyperspectral image (HSI), characterized by hundreds of continuous spectral bands, exhibits distinct advantages in material identification and classification [9]. However, the spectral curves of different land cover types may overlap or exhibit similarities, posing significant challenges for precise land cover classification. Particularly within complex surface coverage areas, disparate land cover categories may exhibit similar spectral curves due to analogous material composition-resulting in limited classification accuracy [10]. With the continuous advancement of multisource RS collaborative observation technologies, the joint utilization of diverse observational data has become increasingly feasible. The mutual complementarity and effective fusion of multisource data facilitate more comprehensive feature extraction, which significantly enhances classification accuracy. Consequently, multisource RS data fusion classification has emerged as a critical research topic [11], [12].

Based on the methodologies in image information processing, multisource RS data fusion techniques can be categorized into three primary levels: pixel-level fusion, feature-level fusion, and decision-level fusion [13], [14], [15]. Due to substantial differences in imaging mechanisms, spatial resolutions, and information representation among various RS data sources, direct pixel-level fusion of raw images often

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

falls short of achieving optimal results [16]. While decision-level fusion integrates information at a higher semantic level, it merely amalgamates classification outcomes at a later stage. Due to its lack of feature interaction depth and reliance on predefined integration rules, its overall effectiveness is diminished [17]. In contrast, feature-level fusion leverages discriminative features extracted from RS data to better exploit the complementary attributes inherent in multisource data demonstrating superior performance in complex land cover classification tasks [18], [19].

In recent years, deep learning-based methods have shown considerable promise in the feature fusion classification of HSI and LiDAR data [20], [21]. Convolutional neural networks (CNNs), characterized by their unique receptive field structures and parameter-sharing convolutional kernels, exhibit robust capabilities for feature extraction, enabling effective integration of HSI and LiDAR data features [22], [23]. Xu et al. [24] designed a dual-branch CNN that extracted spectral and spatial features from HSI incorporating cascade modules within another CNN to derive features from LiDAR data for classification purpose. Fang et al. [25] proposed a spatial–spectral enhancement network (S2ENet) that facilitated information exchange between the features of HSI and LiDAR data within a CNN framework. Roy et al. [26] developed an innovative joint classification method for HSI and LiDAR data utilizing CNNs augmented with spatial morphological blocks to separately extract spectral and spatial feature information. Hong et al. [27] introduced a deep encoder–decoder network architecture (EndNet) that enhanced classification performance by strengthening the fusion of features derived from both HSI and LiDAR datasets.

While CNNs excel in feature extraction, their ability to effectively harness spectral information from HSI remains relatively restricted [28]. To better exploit the spectral information inherent in HSI, attention mechanisms have been integrated into processing models for HSI [29]. By mitigating noise and enhancing focus on prominent features, these mechanisms bolster the model's global modeling capabilities. In recent years, transformer models grounded in self-attention mechanisms have demonstrated remarkable performance in HSI classification [30], [31]. Their robust global modeling capacities and proficiency in capturing long-range dependencies have garnered widespread acclaim within the RS community [32], [33]. Concurrently, with the growing demand for multisource RS data fusion classification, transformer-based classification networks have made significant strides in joint classification of HSI and LiDAR data [34], [35]. Ding et al. [36] designed a global–local transformer network (GLT-Net) that utilized both CNN and transformer architectures to capture intricate spatial–spectral information. Zhao et al. [37] proposed a novel cross-transformer fusion encoder that effectively integrated HSI and LiDAR data, thereby enhancing overall classification performance. He et al. [38] developed a multilevel attention dynamic-scale network (MADNet), incorporating a global–local cross-attention module with distance-weighted operators to effectively minimize redundant information during long-range feature interactions while achieving deep multisource feature integration. Tang et al. [39] proposed a multi-information collaborative fusion network (MICF-Net) that leveraged spatial relationships and high-level semantic information derived from multisource RS data to guide the feature extraction process. Wang et al. [40] introduced a convolutional interactive transformer network (CITNet) that extracted joint local–global features from various sources of RS data through a local–global transformer module (LGTM), facilitating efficient joint classification of HSI and LiDAR data.

Although the aforementioned transformer-based classification methods have demonstrated remarkable performance, existing approaches often remain at shallow fusion levels during cross-modal feature interaction, thus failing to adequately capture the deep semantic relationships between spectral and spatial features of multisource RS data [41], [42]. Furthermore, the global attention mechanisms employed in certain methodologies result in significant computational redundancy within long-range feature interactions, inevitably leading to an increase in the number of parameters and training time for the model [43]. Orthogonal constraints serve as an effective tool for feature decoupling and provide a novel perspective for multisource RS feature fusion [44]. Unlike feature decoupling methods that utilize multibranch network structures or regularization constraints, orthogonal constraints impose mutual independence among feature directions from a vector space perspective. This effectively mitigates feature redundancy. By employing this approach, we eliminate the necessity for complex network modules while preserving model simplicity and computational efficiency, thus achieving rigorous feature decoupling. Enforcing orthogonal constraints ensures that feature vectors are mutually perpendicular in the spatial dimension, thereby reducing the feature redundancy and enhancing independence [45]. The attention mechanism adaptively concentrates on more critical spatial locations. Moreover, by introducing attention-based weight allocation to orthogonalized feature vectors, not only is the flexibility of feature interaction improved but also potential local information loss stemming from orthogonal constraints is mitigated.

In contrast to active RS technologies such as LiDAR, which directly acquires terrain elevation information, both land surface temperature (LST) data derived from thermal infrared data and HSI are classified as the passive RS systems [46], [47]. These systems rely on energy conversion processes involving solar radiation and surface interactions. Specifically, HSI reflects surface characteristics in response to shortwave solar radiation (0.4–2.5 μm), whereas LST data characterize longwave thermal radiation (8–14 μm) emitted following the absorption by the surface [48], [49]. The fusion of these two data types facilitates a more comprehensive analysis of land surface properties. By integrating the spectral features obtained from HSI with the temperature characteristics represented in LST data, a spectral-temperature feature representation can be constructed. This combination provides physically meaningful complementary discriminative information for land cover classification. Consequently, exploring a spatial orthogonal-based method for HSI and LST data joint classification can effectively balance shared and unique features across diverse source of features. This methodology enables deep semantic

alignment while streamlining computational workflows by reducing complex matrix operations. Therefore, it offers an efficient and reliable solution for collaborative classification tasks involving multisource RS data.

Based on the above analysis, this article proposes a fusion network with orthogonality-constrained spatial attention (OCSA-FN) for the joint classification of HSI and LST data. By incorporating orthogonal constraints into the spatial attention mechanism, OCSA-FN effectively mitigates feature redundancy between HSI and LST data while dynamically capturing global dependencies across multisource features. It enhances the network's ability to concentrate on critical information, thereby leveraging the complementary features inherent in both HSI and LST datasets. The main contributions of this study are outlined as follows.

1) A dual branch CNN (DB-CNN) feature extraction module is designed. In this architecture, one branch utilizes a cross spatial–spectral CNN (CSS-CNN) submodule to extract spectral features form HSI; the other branch employs the proposed multiscale learnable Sobel CNN (LS-CNN) submodule to extract temperature features.
2) A dual-pooling residual channel attention (DRCA) module is introduced, which leverages a pooling feature interaction mechanism and residual connect strategy across dual channels to apply channel-wise weighting to the extracted spectral and temperature features. This design enables the network to prioritize feature channels that contain the critical information, thereby effectively minimizing interchannel interference and mitigating feature ambiguity.
3) An adaptive orthogonal feature fusion (AOFF) module is proposed, which integrates orthogonal constraints with a spatial attention mechanism to facilitate efficient interaction and enhancement of multisource features. This module constructs two sets of mutually orthogonal spatial basis vectors within the spatial dimension, effectively reducing redundancy among features. Building upon this foundation, spectral and temperature features are initially projected onto the spatial basis vectors and then adaptively weighted through the application of spatial attention. This dynamic weighting strategy modifies the fusion weights of spatial features between HSI and LST, achieving profound interaction and complementarity between the spectral and temperature features at the level of spatial semantic.

The remainder of this article is organized as follows. Section II provides a comprehensive explanation of the proposed algorithm and details its key components; Section III presents a thorough evaluation of the proposed model through systematic experimentation; the summary and conclusion are displayed in Section IV.

II. Methodology

A. Overall Framework

As depicted in Fig. 1, the proposed OCSA-FN comprises three fundamental components: the DB-CNN, DRCA, and AOFF modules. Within the DB-CNN module, this study employs the CSS-CNN submodule to extract spectral features from HSI, and utilizing the LS-CNN submodule to extract temperature features from LST. Subsequently, the DRCA module assigns weights to both extracted spectral and temperature features, enhancing the expression capability of the key feature channels. Finally, the AOFF module constructs a set of mutually orthogonal basis vectors and integrates them with a spatial attention mechanism to achieve deep interaction between spectral and temperature features.

B. DB-CNN Module

In this part, a DB-CNN module is proposed to effectively address the feature extraction challenges associated with HSI and LST data, which exhibit significant differences in data dimensions within traditional single branch network architectures. The DB-CNN model achieves targeted extraction of multisource features. In particular, a CSS-CNN submodule is designed to harness the rich spectral information inherent in HSI, enabling efficient spectral feature extraction through strategies that fuse multiscale features. Leveraging the spatial gradient characteristics of LST, an LN-CNN submodule is employed for extracting temperature-related features. This dual branch architecture not only effectively reduces data redundancy but also ensures the independence and integrity of multisource features.

1) CSS-CNN Submodule: CNN progressively reduces the spatial dimensions of input data through multilayer convolution operations, thereby enabling effective modeling of image features at various scales and spatial levels. When processing HSIs, 3-D CNN can jointly model both spatial and spectral dimensions, allowing for a comprehensive exploration of the correlation features within his data across these axes. In contrast, 2-D CNN excel in capturing pixel-level surface texture characteristics and local spatial information. Based on this, a CSS-CNN submodule is proposed that synergizes the advantages of both 3-D CNN and 2-D CNN to more comprehensively capture the intricate correlations between spectral and spatial dimensions present in HSI. Furthermore, CSS-CNN submodule utilizes convolution kernels of varying sizes, focusing on different ranges of detail information within HSI. This approach facilitates the development of a hierarchical feature representation system.

The HSI and LST data can be shown as $X_H \in \mathbb{R}^{H\times W\times C}$ and $X_T \in \mathbb{R}^{H\times W}$, where H and W denote the spatial size and C is the number of spectra. The spilt patch cubes are $X_H \in \mathbb{R}^{S\times S\times C}$, where S represents the length and width of the patch. To address the issue of data redundancy, the rincipal component analysis (PCA) [50] is employed for dimensionality reduction, facilitating a preliminary spatial projection of the dataset and decreasing the number of bands to 30. The principal feature components are represented by $X_H^{\mathrm{PCA}} \in \mathbb{R}^{S\times S\times D}$, and D represents the number of bands after dimension reduction. In the CSS-CNN sub-module, a 3-D CNN with a kernel size of $3\times 3\times 3$ is employed, followed by three parallel branches of 2-D CNNs featuring kernel sizes of 3×3, 5×5, and 7×7, respectively. During the convolution operations, zero-padding is applied to ensure consistent spatial dimensions between the output spectral features and the input imagery.

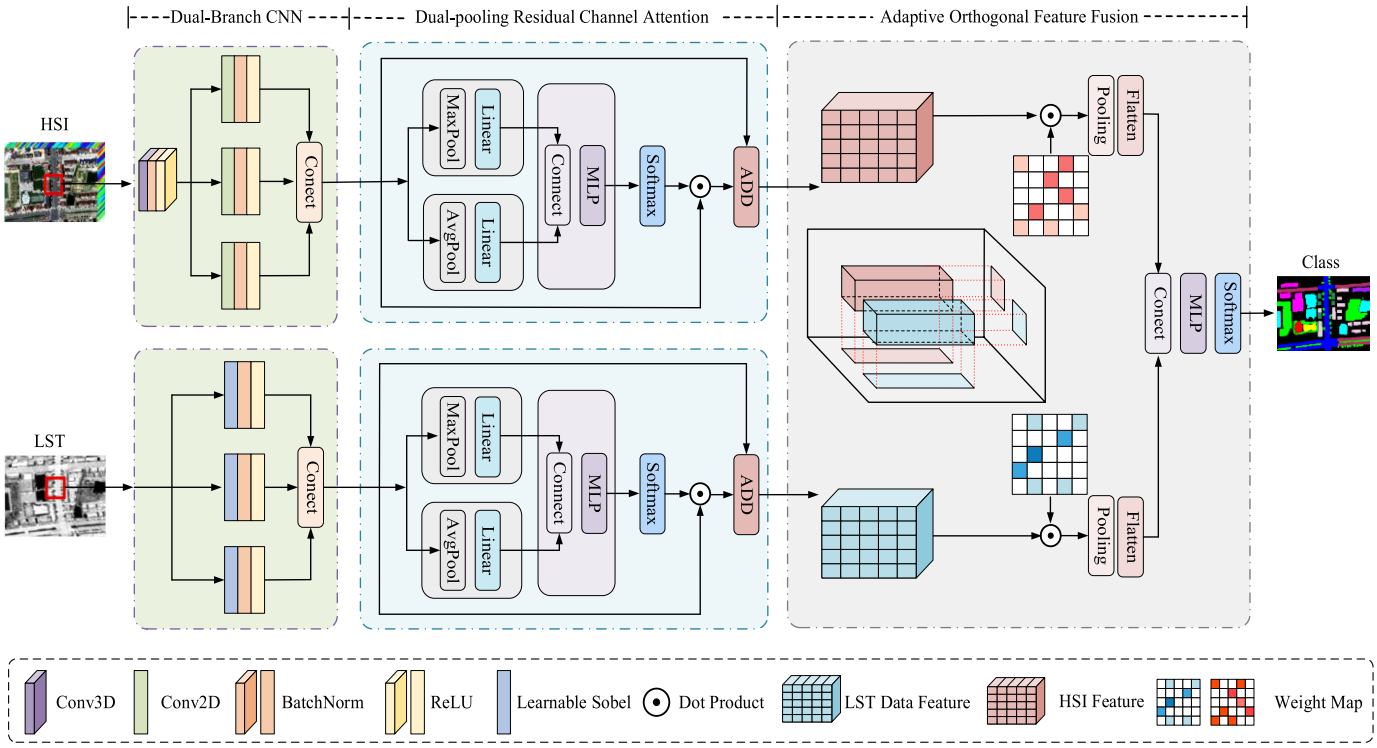


Fig. 1. Illustration of the proposed OCSA-FN.

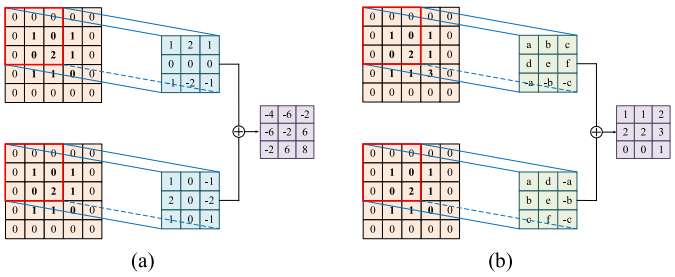


Fig. 2. Comparison of feature extraction methods. (a) Calculate using the traditional Sobel operator. (b) Perform calculations using LS-CNN.

Each convolutional layer is sequentially connected with batch normalization (BN) and rectified linear unit (ReLU) activation layers. Ultimately, the extracted spectral features are concatenated to form the HSI feature representation $F_H \in \mathbb{R}^{S\times S\times N}$, where N denotes the number of feature channels

$$X'_H = \varphi_{3\times3\times3}\left(X_H^{\mathrm{PCA}}\right) \tag{1}$$

$$F_H = \varphi_{3\times3}\left(X'_H\right) \oplus \varphi_{5\times5}\left(X'_H\right) \oplus \varphi_{7\times7}\left(X'_H\right) \tag{2}$$

where $\varphi_{3\times3\times3}$ represents the 3-D CNN with a convolution kernel of $3\times3\times3$. $\varphi_{3\times3}$, $\varphi_{5\times5}$, and $\varphi_{7\times7}$ represent the 2-D CNN with convolution kernel sizes of 3×3, 5×5, and 7×7, respectively. $\oplus$ represents the connect operation.

2) LS-CNN Submodule: The traditional Sobel operator serves as a foundational tool for edge feature extraction, primarily designed to identify edge feature by calculating gradient magnitudes at each pixel. This operator typically employs two fixed 3×3 convolutional kernels for horizontal and vertical edge detections, respectively, as illustrated in Fig. 2(a). The final edge results are derived from the summation of both directional outputs. However, the Sobel operator is hindered by two fundamental limitations: 1) its exclusive focus on edge features fails to capture spatial information in nonedge regions and 2) the detection accuracy is heavily reliant on the empirically set thresholds, potentially leading to suboptimal edge localization when inappropriate threshold selection occurs.

To address these limitations, this part introduces a LS-CNN submodule specifically designed for the extraction of temperature feature from LST data. This architecture synergistically integrates the data distribution characteristics of traditional Sobel operators with the learnable kernels of CNN as illustrated in Fig. 2(b). Different from the traditional Sobel operator, which is primarily constrained to edge features extraction, the LS-CNN adaptively learns more comprehensive feature representations that encompass nonedge features. This is accomplished by incorporating learnable nonzero parameters within the central rows and columns of its convolutional kernels. Such a trainable mechanism enables the LS-CNN to address and transcend the functional limitations inherent in the Sobel operator.

The learnable 3×3 convolutional kernel W_{LS} in the LS-CNN submodule is formally defined as

$$W_{\mathrm{LS}} = \begin{pmatrix} a & d & -a \\ b & e & -b \\ c & f & -c \end{pmatrix} \tag{3}$$

where a, b, c, d, e, and f are the learnable parameters.

Regarding the LST data patch cubes as $X_T \in \mathbb{R}^{S\times S}$, the submodule implements LS-CNN convolution operations in both horizontal and vertical directions. The results from

these two orientations are then aggregated to obtain the corresponding temperature features. Furthermore, the module adopts a multiscale LS-CNN architecture that utilizes three distinct convolutional kernel dimensions: 3×3, 5×5, and 7×7. This design enables the simultaneous detection of fine edges as well as large contours, thus facilitating more comprehensive extraction of both edge and nonedge features from LST data. Subsequently, a BN layer is incorporated to scale the features, aiming to stabilize the training process of the model and accelerate convergence. Following this step, the ReLU is employed as the activation function to further enhance the nonlinear representation capacity of the extracted features. The specific formula for calculating the temperature feature $F_T \in \mathbb{R}^{S\times S\times N}$ is presented as follows:

$$\phi_{3\times3} = W_{\mathrm{LS}} \times X_T + W_{\mathrm{LS}}^T \times X_T \tag{4}$$

$$F_T = \phi_{3\times3}(X_T) \oplus \phi_{5\times5}(X_T) \oplus \phi_{7\times7}(X_T) \tag{5}$$

where $\phi_{3\times3}$, $\phi_{5\times5}$, and $\phi_{7\times7}$ represent the LS-CNN with convolution kernel sizes of 3×3, 5×5, and 7×7 in the horizontal and vertical directions, respectively. $\times$ represents convolution operation.

C. DRCA Module

A DRCA module is designed to perform channelwise feature weighting on the extracted spectral feature F_H and temperature feature F_T. The DRCA dynamically adjusts channel weights by assessing the significance of each feature channel. Furthermore, it employs residual connections to ensure effective propagation of critical feature information throughout the network, thereby enhancing the model's capability to capture and leverage salient features from the input data.

The DRCA module first applies both global max pooling and global average pooling along the spatial dimensions of F_H and F_T, generating channelwise attention scores. These scores are then processed through a linear layer, concatenated via a connect function, and transformed nonlinearly using an MLP. Finally, the Softmax function is utilized to convert the MLP outputs into attention weights, which are applied to each channel to produce weighted features $F'_H \in \mathbb{R}^{S\times S\times N}$ and $F'_T \in \mathbb{R}^{S\times S\times N}$. To enhance the feature propagation stability and mitigate gradient vanishing, the module performs elementwise addition between the original input features F_H and F_T and their corresponding weighted features F'_H and F'_T, yielding the final residual-enhanced channel-weighted features $F_H^C \in \mathbb{R}^{S\times S\times N}$ and $F_T^C \in \mathbb{R}^{S\times S\times N}$. This process can be formally expressed as follows:

$$F'_H = \sigma\left(\mathrm{MLP}\left((\mathrm{Avg}(F_H)) \oplus (\mathrm{Max}(F_H))\right)\right) \cdot F_H \tag{6}$$

$$F_H^C = F_H + F'_H \tag{7}$$

$$F'_T = \sigma\left(\mathrm{MLP}\left((\mathrm{Avg}(F_T)) \oplus (\mathrm{Max}(F_T))\right)\right) \cdot F_T \tag{8}$$

$$F_T^C = F_T + F'_T \tag{9}$$

where σ represents the Softmax function. Avg and Max represent the global average pooling and global max pooling, respectively.

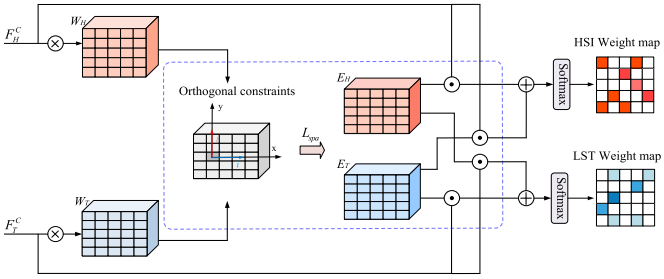


Fig. 3. Architecture of proposed AOFF module.

D. AOFF Module

The proposed architecture incorporates a AOFF module designed to facilitate deep interaction between spectral and temperature features, as illustrated in Fig. 3. This innovative AOFF module synergistically integrates orthogonal constraints with spatial attention mechanisms to achieve comprehensive feature fusion.

The spectral-weighted feature F_H^C and temperature-weighted feature F_T^C, processed by the DRCA module, undergo linear projection via two parameter matrices $W_H \in \mathbb{R}^{S\times S\times N}$ and $W_T \in \mathbb{R}^{S\times S\times N}$ of identical dimensions, yielding transformed feature matrices $E_H \in \mathbb{R}^{S\times S\times N}$ and $E_T \in \mathbb{R}^{S\times S\times N}$ as formalized in (10) and (11). To enable a comprehensive interaction between spectral and temperature features, this part proposes a feature optimization method based on spatial orthogonal constraints. Specifically, for each spatial location (i, j) in feature matrices E_H and E_T, the channel vectors $e_{H(i,j)} \in \mathbb{R}^N$ and $e_{T(i,j)} \in \mathbb{R}^N$ are extracted and subjected to orthogonal constraints to ensure their orthogonality in the feature space. Consequently, the feature matrices E_H and E_T collectively form a set of orthogonal basis vectors within the spatial dimension

$$E_H = F_H^C \cdot W_H \tag{10}$$

$$E_T = F_T^C \cdot W_T. \tag{11}$$

At each spatial location (i, j), the channel vectors $f_{H(i,j)}^C \in \mathbb{R}^N$ and $f_{T(i,j)}^C \in \mathbb{R}^N$ are extracted from the spectral feature F_H^C and temperature feature F_T^C, respectively. These vectors are subsequently projected onto their corresponding positional feature vectors $e_{H(i,j)}$ and $e_{T(i,j)}$ to generate spatial attention scores. The combined attention scores from both modalities are then summed and normalized via the Softmax function to yield the spatial attention weights. Ultimately, these weights are applied to weight the original HSI feature F_H^C and LST feature F_T^C. This process can be formally expressed as follows:

$$F_H^S = \sigma\left(F_H^C \cdot E_H + F_H^C \cdot E_T\right) \cdot F_H^C \tag{12}$$

$$F_T^S = \sigma\left(F_T^C \cdot E_H + F_T^C \cdot E_T\right) \cdot F_T^C \tag{13}$$

where $F_H^S \in \mathbb{R}^{S\times S\times N}$ and $F_T^S \in \mathbb{R}^{S\times S\times N}$ represent the spatial attention weighted HSI feature and LST feature, respectively.

The projection and spatial weighting of feature matrices F_H^C and F_T^C onto E_H and E_T are computed independently to more effectively capture the distinctiveness of each feature representation. Throughout the spatial weighting process, the aggregation of attention scores amplifies the feature responses

at critical spatial locations, thereby promoting a profound interaction between the two features.

E. Joint Loss Function

This part proposes a joint loss function optimization strategy that guides model training by integrating spatial orthogonal loss L_{spa} and classification prediction loss L_{cls}. The final composite loss L is formulated as a weighted summation of these two objective functions

$$L = L_{\text{cls}} + \lambda L_{\text{spa}} \tag{14}$$

where λ is the weight coefficient of spatial orthogonality loss, which is used to balance the relationship between classification accuracy and feature spatial orthogonality. As an important hyperparameter, the value of λ is further determined through systematic parameter analysis experiments.

L_{spa} is specifically designed to impose orthogonal constraints between feature vectors. This loss function quantifies the degree of the orthogonal deviation among pairs of feature vectors across all spatial positions. When L_{spa} approaches 0, the resulting feature matrices E_H and E_T form a set of orthogonal basis vectors along the spatial dimension. The formulation of L_{spa} is given as follows:

$$L_{\text{spa}} = \frac{1}{S \times S} \sum_{i=1}^{S} \sum_{j=1}^{S} \left(e_{H(i,j)} \cdot e_{T(i,j)} \right)^2. \tag{15}$$

For the classification loss, spectral feature F_H^S, and temperature feature F_T^S, which are processed by the DRCA and AOFF modules, undergo pooling and flattening operations prior to being fused through a connect function. The combined features are subsequently input into the MLP, where a Softmax function output layer generates the classification probabilities.

L_{cls} quantifies the discrepancy between the predicted probability distribution and the ground-truth labels, formally expressed as

$$\hat{y} = \sigma\left(\text{MLP}\left(\text{Flat}\left(\text{Avg}\left(F_H^S\right)\right) \oplus \text{Flat}\left(\text{Avg}\left(F_T^S\right)\right)\right)\right) \tag{16}$$

$$L_{\text{cls}} = -\frac{1}{M} \sum_{i=1}^{M} y_i^T \log\left(\hat{y}_i\right) \tag{17}$$

where Flat represents the flattening operation, M represents the number of categories in the classification task, y_i^T represents the true label of the sample, and $\hat{y}_i$ is the predicted probability distribution of the model.

III. Experimental Results and Discussion

A. Dataset Introduction

The dataset utilized in this study comprises aerial HSI and thermal infrared hyperspectral imagery (TI-HSI) acquired simultaneously over Hengdian Town, Dongyang City, Zhejiang Province, China. Data collection occurred concurrently at 13:15 local time on May 24, 2019. The HSI data consist of 230 spectral channels covering the wavelength range from 0.437 to 0.985 μm, with a spectral resolution of 2.38 nm and a spatial resolution of 1 m. The TI-HSI data includes 110 spectral bands spanning from 8.06 to 11.22 μm, with a spatial resolution of 2 m and a spectral resolution of 28.7 nm. The LST is derived from the TI-HSI data through the temperature-emissivity separation (TES) algorithm [51], which facilitates simultaneous retrieval of both temperature and emissivity at the pixel level.

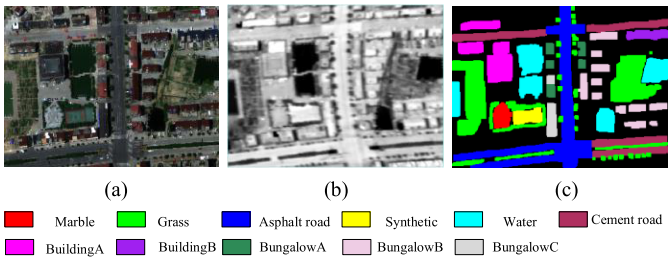


Fig. 4. Urban dataset. (a) False-color image of HSI. (b) LST data map. (c) Ground truth map.

TABLE I
Number of Training and Test Samples for the Urban Dataset

Class	Class Name	Training	Test
1	Marble	9	911
2	Grass	118	11,745
3	Asphalt road	94	9,311
4	Synthetic	9	947
5	Water	57	5,705
6	Cement road	55	5,500
7	BuildingA	37	3,663
8	BuildingB	8	888
9	BungalowA	11	1,123
10	BungalowB	30	3,056
11	BungalowC	9	969
-	Total	437	43,818

Following the preprocessing phase, this study constructs three classification datasets: an Urban dataset, a Lake dataset, and a Bridge dataset. To evaluate the effectiveness of the proposed OCSA-FN model, this study applies the model to these three generated datasets, designating 1% of the data allocated as the training set while the remaining 99% as the test set.

1) Urban Dataset: The Urban dataset primarily comprises land cover types including buildings, roads, water bodies, and grasslands. It has a spatial dimension of 251 × 334 pixels, yielding a total of 44 255 sampled pixels across 11 test categories. Fig. 4 presents the false-color composite of the HSI, the LST data map, and the ground truth map of the Urban dataset, respectively. Table I details the categorywise distribution of training and test samples used in the experiments.

2) Lake Dataset: The Lake dataset mainly consists of land cover types such as lake, grasslands, and buildings, comprising 272 × 423 pixels with a total of 72 976 sample pixels covering 16 test categories. This dataset is dominated by large lake water bodies while also including grasslands and a small number of man-made structures. Fig. 5 illustrates the visualization results of the Lake dataset, including the false-color HSI, the LST data map, and the ground truth map. Table II provides the detailed class distribution of training and test samples.

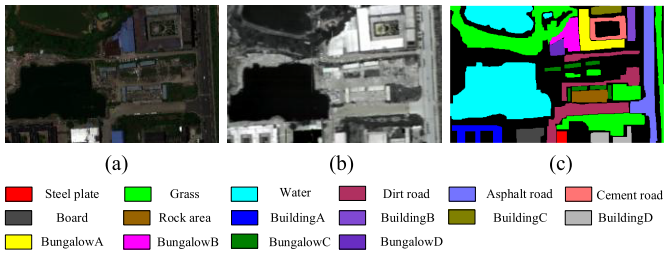


Fig. 5. Lake dataset. (a) False-color image of HSI. (b) LST data map. (c) Ground truth map.

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR THE LAKE DATASET

Class	Class Name	Training	Test
1	Steel plate	4	456
2	Grass	184	18,254
3	Water	234	23,210
4	Dirt road	73	7,277
5	Asphalt road	58	5,759
6	Cement road	16	1,623
7	Board	14	1,455
8	Rock area	16	1,619
9	BuildingA	21	2,123
10	BuildingB	10	1,033
11	BuildingC	9	926
12	BuildingD	13	1,353
13	BungalowA	28	2,772
14	BungalowB	21	2,108
15	BungalowC	15	1,518
16	BungalowD	7	767
-	Total	723	72,253

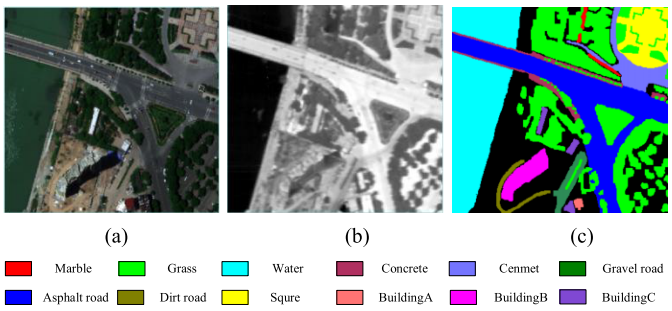


Fig. 6. Bridge dataset. (a) False-color image of HSI. (b) LST data map. (c) Ground truth map.

3) Bridge Dataset: The bridge dataset encompasses land cover types such as bridge surfaces, highways, water bodies, and grasslands, with a spatial size of 283 × 307 pixels. It contains 65 744 valid samples spanning 12 land cover categories, including buildings, grasslands, water bodies, and roads. The specific classes and their sample quantities are listed in Table III. Fig. 6 displays the visualization results of the bridge dataset, including the false-color HSI, the LST data map, and the ground truth map.

B. Experimental Preparation

1) Experimental Platform: This study conducts all experiments on a hardware platform with an NVIDIA GeForce RTX 3090 GPU, Intel Core i9-10900K CPU, and 30 GB RAM. The implementation uses the PyTorch 1.10.0 framework with Python 3.8 under Windows 11. The model employs the Adam algorithm to optimize all learnable parameters, with the number of training epochs set to 500, batch size to 64, and learning rate to 0.001.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR THE BRIDGE DATASET

Class	Class Name	Training	Test
1	Marble	4	454
2	Grass	208	20,628
3	Water	167	16,567
4	Concrete	18	1,875
5	Cement	29	2,954
6	Gravel road	12	1,217
7	Asphalt road	138	13,745
8	Dirt road	8	823
9	Square	41	4,120
10	BuildingA	2	162
11	BuildingB	19	1,891
12	BuildingC	6	656
-	Total	652	65,092

2) Evaluation Indicators: To assess the classification performance of the proposed method as well as the comparative models, four widely used metrics are adopted: overall accuracy (OA), average accuracy (AA), Kappa coefficient, and per-class precision. For all these metrics, higher values denote superior classification performance. To ensure the reliability of experimental outcomes, the mean values from multiple trials are presented as the final metrics.

C. Parameter Analysis Experiment

1) Input Patch Size: Patches of different sizes can capture spatial features of ground objects at varying scales, thereby influencing the classification performance of the network through spatial contextual information. This study presents a series of systematic experiments conducted across the three datasets with patch sizes of {7, 9, 11, 13, 15, 17} to assess their impact on classification accuracy. As demonstrated by the experimental results shown in Fig. 7, the OCSA-FN achieves the highest OA on both Urban and Bridge datasets when the patch size is set to 13. For the Lake dataset, the optimal performance is observed with a patch size of 15.

2) Learning Rate: The learning rate is a crucial hyperparameter in deep learning, significantly influencing both the training process and classification performance. A higher learning rate can expedite model convergence; however, it may also cause divergence from optimal solutions. Conversely, a lower learning rate promotes more stable training but poses the risk of trapping network parameters in local minima, thereby complicating the attainment of global optima. To assess the effect of learning rate on model performance, this study evaluates six different values: $\{5 \times 10^{-2}, 1 \times 10^{-2}, 5 \times 10^{-3}, 1 \times 10^{-3}, 5 \times 10^{-4}, 1 \times 10^{-4}\}$ across all three datasets. As evidenced in Fig. 8, experimental results demonstrate that OCSA-FN achieve optimal classification performance across all datasets when the learning rate is set to 1×10^{-3}.

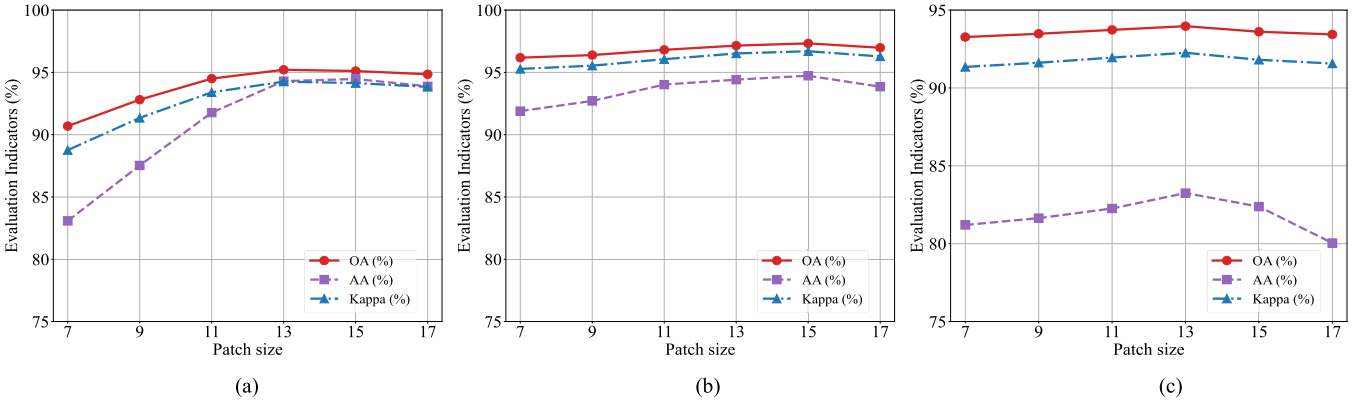


Fig. 7. Impact of patch size on classification performance of different datasets. (a) Urban dataset. (b) Lake dataset. (c) Bridge dataset.

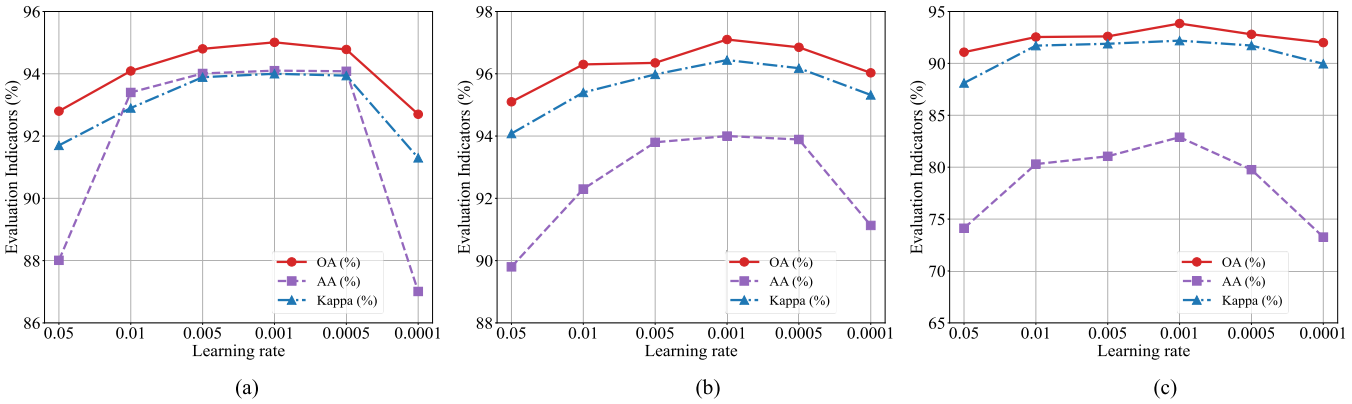


Fig. 8. Influence of learning rate on classification performance of different datasets. (a) Urban dataset. (b) Lake dataset. (c) Bridge dataset.

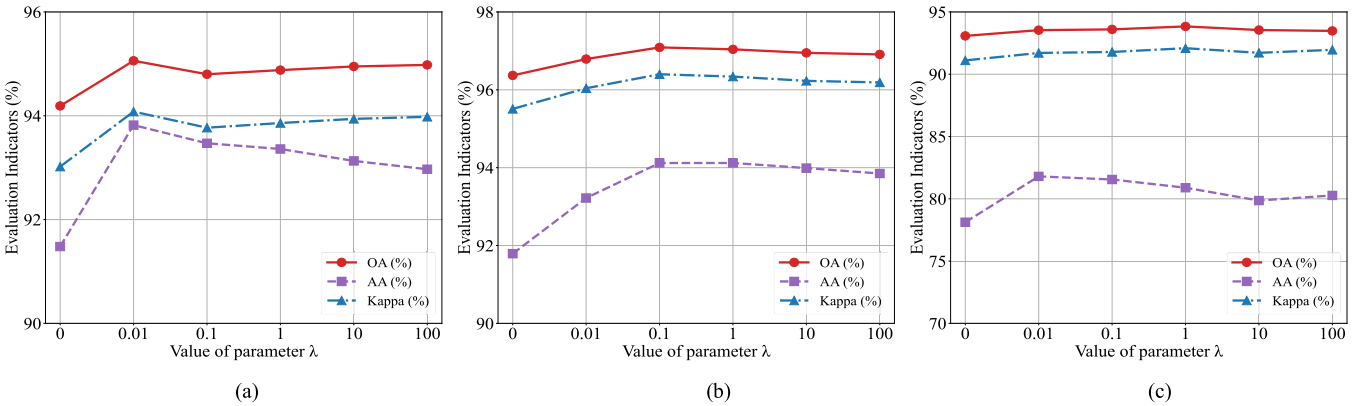


Fig. 9. Impact of λ on classification performance across different datasets. (a) Urban dataset. (b) Lake dataset. (c) Bridge dataset.

3) Spatial Orthogonal Loss Weight: The hyperparameter λ in (14) regulates the strength of the orthogonal constraints between spectral and temperature features in the spatial dimension. Proper optimization of λ significantly enhances multisource information fusion, leading to improved classification accuracy. Fig. 9 presents the experimental results with different values of λ {0.01, 0.1, 1, 10, 100} on the three datasets. The findings demonstrate that different datasets exhibit varying sensitivities to λ: the OCSA-FN attains optimal performance on the Urban dataset, Lake dataset, and Bridge dataset when λ is set to 0.01, 0.1, and 1, respectively. Notably, a value of $\lambda = 0$ leads to a substantial decline in classification performance across all three datasets. This phenomenon indicates that the orthogonal constraints in the spatial dimension can effectively reduce feature redundancy while enhancing feature independence and representational capability, ultimately improving the model's classification performance.

D. Comparative Experiment

To evaluate the performance of the proposed OCSA-FN model, this study conducts comparative experiments with

TABLE IV
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON URBAN DATASET

Class	CCR-Net	EndNet	S2ENet	MDL	Coupled CNN	ExVit	HCT	MACN	GLT-Net	MASS Former	OCSA-FN
1	65.54	55.33	93.03	91.56	93.15	96.20	81.41	96.62	83.58	96.41	**96.63**
2	90.38	88.31	94.74	94.06	**94.89**	92.27	93.12	92.00	93.11	93.26	94.58
3	88.48	85.18	92.15	92.65	97.23	97.78	95.78	96.28	**98.68**	96.21	96.60
4	79.08	80.13	96.10	95.17	96.44	97.49	99.58	69.87	93.93	98.64	**99.68**
5	97.94	86.69	94.35	94.17	97.27	98.07	97.62	98.05	**98.76**	98.47	97.30
6	81.24	85.42	92.68	92.08	90.44	92.66	92.90	93.10	92.02	86.17	**93.68**
7	61.30	65.03	94.01	**97.57**	93.10	91.78	93.18	93.05	96.51	96.97	94.18
8	19.53	67.19	93.71	92.72	91.40	97.99	**100.00**	99.66	88.39	85.48	93.75
9	69.40	79.28	96.17	96.74	96.47	99.03	98.58	**99.11**	94.88	94.80	96.91
10	84.02	79.16	97.57	97.74	97.05	98.48	99.04	98.60	97.89	97.18	**99.06**
11	41.62	44.89	56.91	88.04	80.47	69.84	86.29	85.07	72.49	**94.33**	79.95
OA	83.13	82.00	91.44	91.98	94.79	94.62	94.70	94.19	94.83	94.88	**95.27**
AA	70.78	74.24	88.53	91.96	93.45	93.78	94.34	92.86	91.84	94.35	**94.58**
Kappa	78.80	78.45	89.84	90.49	93.77	93.57	93.66	93.05	93.81	93.63	**94.33**

TABLE V
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON LAKE DATASET

Class	CCR-Net	EndNet	S2ENet	MDL	Coupled CNN	ExVit	HCT	MACN	GLT-Net	MASS Former	OCSA-FN
1	81.95	93.91	47.17	72.61	99.78	96.74	**100.00**	**100.00**	92.17	96.30	97.17
2	96.29	95.25	89.61	89.51	99.12	97.89	99.22	96.83	98.79	97.56	**99.34**
3	98.73	97.07	95.86	95.80	99.55	99.60	99.50	98.86	99.46	98.82	**99.95**
4	93.65	89.32	96.26	96.16	92.69	95.09	94.73	97.19	**98.35**	96.68	96.25
5	89.80	81.61	95.53	95.65	98.02	**98.85**	97.91	97.28	97.83	94.82	96.52
6	21.29	21.05	**88.29**	80.17	75.96	71.51	64.12	54.54	69.24	72.94	75.14
7	3.97	90.61	80.60	80.60	98.97	**99.80**	99.79	99.72	99.79	88.93	99.14
8	2.96	30.09	90.95	87.89	85.32	81.60	33.08	51.98	75.96	**99.80**	82.41
9	88.24	81.34	86.33	86.66	99.11	98.83	93.65	98.92	96.87	98.60	**99.98**
10	67.11	55.23	93.58	94.73	94.15	98.66	98.27	97.41	94.15	98.95	**99.66**
11	98.28	90.05	70.27	70.91	98.50	99.89	**100.00**	**100.00**	98.82	**100.00**	99.73
12	28.89	74.67	66.25	58.57	86.53	93.12	92.97	99.26	91.14	**99.78**	96.66
13	99.28	92.32	99.46	99.98	98.71	98.89	99.53	99.46	99.89	99.64	**100.00**
14	83.03	80.23	88.35	89.20	88.68	87.83	85.86	**89.57**	88.77	85.74	89.17
15	52.51	91.78	90.74	92.11	**96.02**	89.43	94.65	85.90	87.47	84.37	87.21
16	30.74	81.01	98.06	97.29	95.73	97.80	**99.61**	97.93	98.70	97.42	98.83
OA	88.46	88.71	92.00	91.86	97.02	96.85	95.75	95.50	96.94	97.16	**97.41**
AA	69.66	77.85	86.08	86.74	94.19	94.10	90.81	91.55	92.97	94.40	**94.81**
Kappa	85.74	86.06	90.31	90.08	96.32	96.11	94.75	94.46	96.22	96.50	**96.80**

TABLE VI
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON BRIDGE DATASET

Class	CCR-Net	EndNet	S2ENet	MDL	Coupled CNN	ExVit	HCT	MACN	GLT-Net	MASS Former	OCSA-FN
1	12.66	29.91	49.56	68.34	**78.38**	54.80	51.09	47.37	50.65	42.79	49.11
2	89.58	93.77	91.55	94.75	96.00	**96.50**	95.42	93.33	94.49	96.39	94.59
3	59.53	92.40	89.50	87.94	97.28	99.06	97.95	97.60	99.08	88.66	**99.40**
4	26.65	33.12	73.72	42.57	62.07	71.90	55.89	67.45	62.33	**88.50**	56.80
5	36.40	70.69	**88.97**	69.79	81.93	83.34	86.55	85.14	86.95	83.08	85.04
6	67.01	87.06	**99.58**	84.03	92.96	91.23	94.20	93.74	94.90	99.42	96.25
7	41.88	54.15	63.54	50.90	84.59	94.71	80.62	87.36	**96.51**	70.16	86.92
8	29.21	10.69	**96.75**	70.87	82.34	68.19	94.38	92.18	64.92	91.05	93.07
9	32.17	10.56	33.81	18.23	92.83	**97.91**	83.70	90.69	96.08	76.60	92.56
10	59.15	53.04	83.54	26.83	21.34	48.17	**100.00**	81.09	48.78	87.20	84.55
11	95.94	92.89	94.43	94.78	98.32	77.23	99.52	99.28	92.46	92.11	**99.63**
12	97.31	98.67	98.51	97.83	89.72	73.87	80.06	92.59	68.42	**99.47**	82.59
OA	83.62	86.39	91.27	88.63	93.17	93.12	92.99	92.95	93.34	93.53	**93.98**
AA	53.96	60.17	80.29	67.24	81.48	79.82	84.95	85.28	79.41	84.62	**85.35**
Kappa	78.79	82.38	88.74	87.83	91.24	91.18	90.98	90.76	91.48	91.73	**92.26**

several representative deep learning-based data fusion classification methods. The comparison methods are categorized into two groups: CNN-based and transformer-based approaches. The CNN-based methods include CCR-Net [52], EndNet [27], S2ENet [25], MDL [53], and CoupledCNN [54], whereas the transformer-based methods comprise ExVit [55],

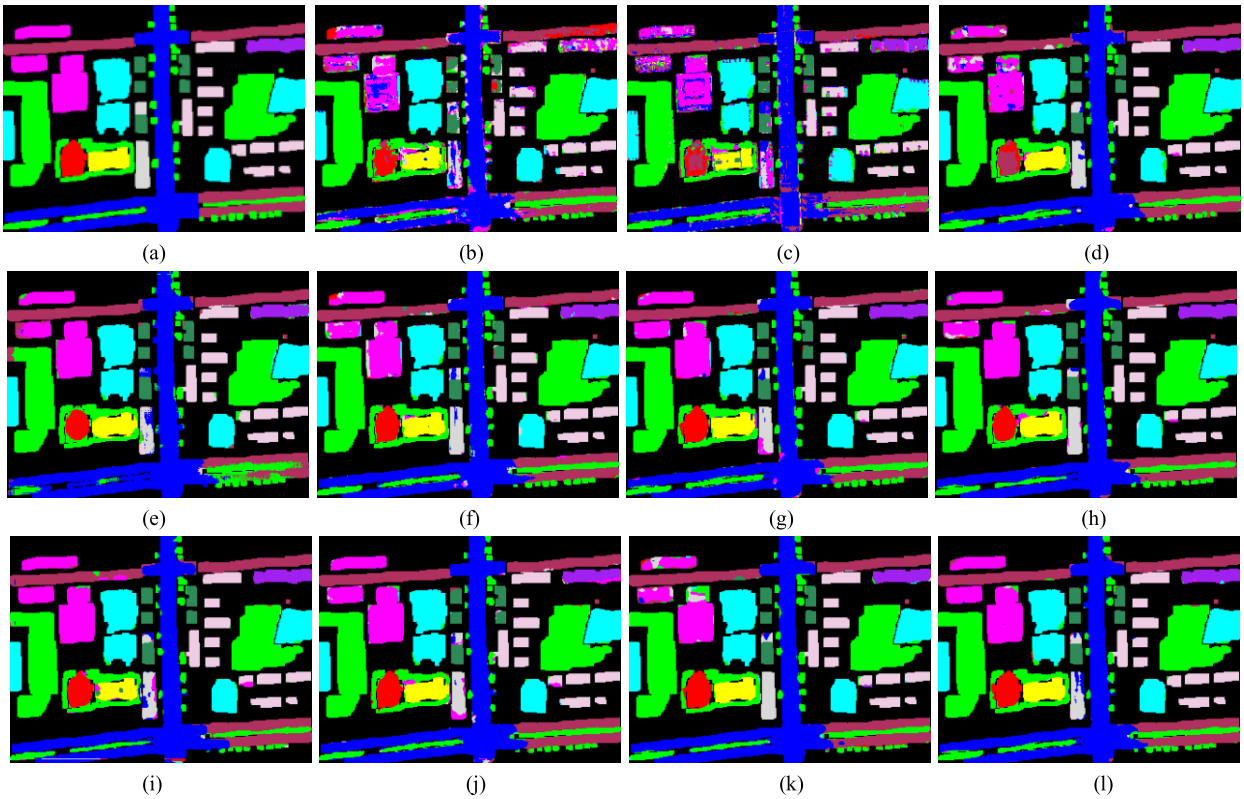


Fig. 10. Comparative analysis of classification maps for the urban dataset using different method. (a) Ground truth. (b) CCR-Net. (c) EndNet. (d) S2ENet (e) MDL. (f) CoupledCNN. (g) ExVit. (h) HCT. (i) MACN. (j) GLT-Net. (k) MASSFormer. (l) OCSA-FN.

HCT [37], MACN [56], GLT-Net [36], and MASSFormer [57]. All comparison methods are implemented utilizing their publicly available codes with default parameter settings. To ensure the reliability of experimental results, each method is independently executed 10 times, with the final results representing the average of these runs. Tables IV–VI present the classification accuracies and per-class accuracy rates for different methods across the three datasets, where the optimal values are highlighted in bold. The experimental findings demonstrate that OCSA-FN achieves superior classification performance on all three datasets. Notably, the encoder–decoder structured EndNet and cross-modal reconstruction strategy-based CCR-Net exhibit relatively poor performance across all datasets. In contrast, S2ENet with the cross-modal interaction enhancement mechanism and CoupledCNN with the coupled CNN architecture demonstrate significant advantages. This indicates that enhancing multisource feature interactions can effectively improve the model performance. Such improvements are particularly pronounced when dealing with classes characterized by a limited number of training samples.

Notably, the five transformer-based classification methods exhibit substantial performance variations. The GLT-Net model, which employs global–local feature representation through multiscale information fusion and image reconstruction tasks, outperformed the HCT model that combines hierarchical CNN and transformer branches. In comparison to the ExVit model, which extends traditional vision transformer, the MACN model with the hybrid self-attention and CNN architecture shows slightly inferior OA. This limitation is predominantly attributed to its limited spectral–spatial interaction during feature fusion. Among these models, MASSFormer stands out with the memory-augmented spatial–spectral attention by incorporating memory tokens to enhance the conventional multihead self-attention operation for comprehensive information fusion.

Compared to the top-performing CNN-based method CoupledCNN and transformer-based approach MASSFormer, the proposed OCSA-FN demonstrates improvements in OA by 0.3%, 0.3%, and 0.5% on the three datasets, respectively. Furthermore, OCSA-FN achieves near or fully reaches the highest per-class accuracy among all compared methods, attributable to its effective cross-modal information interaction. In addition to quantitative analysis, this study provides visual validation of OCSA-FN's effectiveness through Figs. 10–12, which illustrate classification results across various method on the three datasets.

E. Ablation Experiment

To comprehensively evaluate the contributions of each module in OCSA-FN toward improving classification performance, extensive ablation experiments are conducted across the three datasets (as displayed in Table VII). For the LST branch, the proposed LS-CNN module is compared against traditional methods based on Sobel operators. The LS-CNN module demonstrates improvements in OA of 0.7%, 1.0%, and 0.4% across three datasets, respectively, when compared to the traditional Sobel operator, demonstrating its superior capability

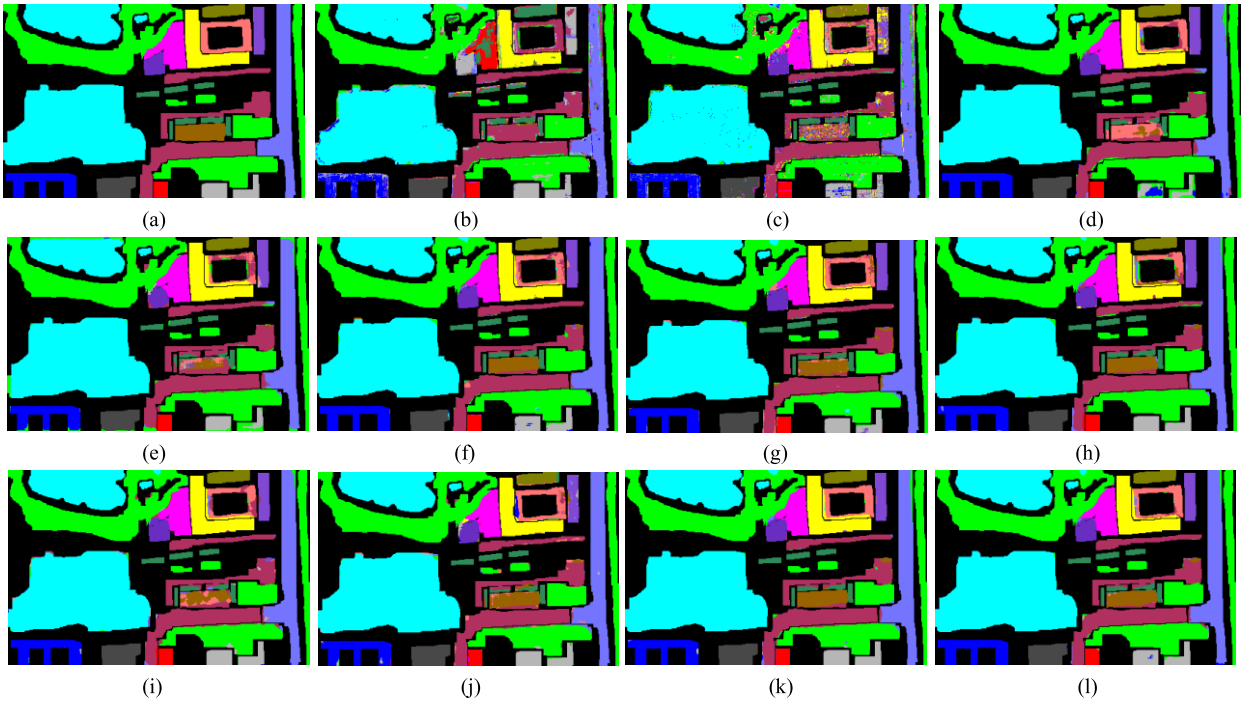


Fig. 11. Comparative analysis of classification maps for the lake dataset using different method. (a) Ground truth. (b) CCR-Net. (c) EndNet. (d) S2ENet (e) MDL. (f) CoupledCNN. (g) ExVit. (h) HCT. (i) MACN. (j) GLT-Net. (k) MASSFormer. (l) OCSA-FN.

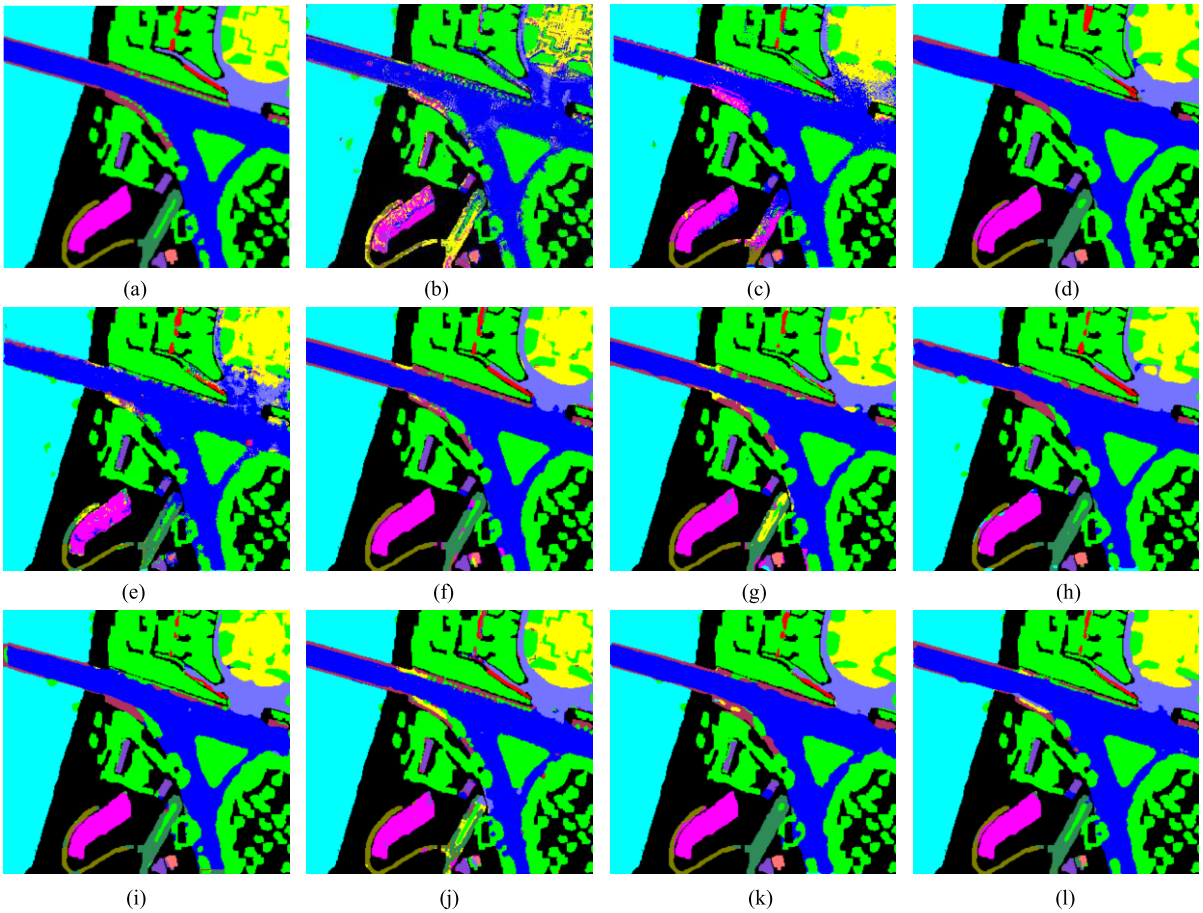


Fig. 12. Comparative analysis of classification maps for the bridge dataset using different method. (a) Ground truth. (b) CCR-Net. (c) EndNet. (d) S2ENet. (e) MDL. (f) CoupledCNN. (g) ExVit. (h) HCT. (i) MACN. (j) GLT-Net. (k) MASSFormer. (l) OCSA-FN.

for feature extraction. In relation to the HSI branch, the CSS-CNN module, which incorporates 3-D and 2-D CNNs, is benchmarked against models that employed either solely 3-D or exclusively 2-D CNNs. The results show that the

TABLE VII
ABLATION STUDY FOR URBAN, LAKE, AND BRIDGE DATASET

Dataset	LST Branch		HSI Branch			Pooling strategy			AOFF	OA	AA	Kappa
	Sobel	LS-CNN	3D-CNN	2D-CNN	CSS-CNN	MaxPool	AvgPool	DRCA				
Urban	✓				✓			✓	✓	94.53	92.81	93.44
		✓	✓					✓	✓	93.45	89.45	92.14
		✓		✓				✓	✓	91.99	88.37	90.39
		✓			✓	✓			✓	94.69	93.24	93.63
		✓			✓		✓		✓	94.39	92.64	93.26
		✓			✓				✓	94.26	91.22	93.11
		✓			✓			✓		88.29	76.65	85.86
		✓			✓			✓	✓	**95.27**	**94.58**	**94.33**
Lake	✓				✓			✓	✓	96.25	91.41	95.35
		✓	✓					✓	✓	95.33	89.29	94.22
		✓		✓				✓	✓	93.96	85.09	92.53
		✓			✓	✓			✓	96.26	92.05	95.38
		✓			✓		✓		✓	96.19	92.25	95.29
		✓			✓				✓	95.75	92.09	95.18
		✓			✓			✓		93.34	80.99	91.76
		✓			✓			✓	✓	**97.27**	**94.05**	**96.63**
Bridge	✓				✓			✓	✓	93.53	80.80	91.69
		✓	✓					✓	✓	92.85	78.95	90.83
		✓		✓				✓	✓	92.11	73.20	89.89
		✓			✓	✓			✓	93.60	80.89	91.73
		✓			✓		✓		✓	93.46	81.32	91.62
		✓			✓				✓	92.88	79.01	90.64
		✓			✓			✓		89.50	62.52	86.45
		✓			✓			✓	✓	**93.97**	**83.84**	**92.27**

TABLE VIII
COMPARISON OF THE NUMBER OF PARAMETERS, FLOPS, AND TRAINING/TEST TIME

Dataset	Complexity	CCR-Net	EndNet	S2ENet	MDL	Coupled CNN	ExVit	HCT	MACN	GLT-Net	MASS Former	OCSA-FN
Urban	Parameters (M)	0.05	0.09	0.37	0.13	0.10	0.23	0.43	0.16	0.73	0.41	0.11
	FLOPs (G)	0.11	0.01	4.01	0.58	0.24	2.95	1.19	1.01	14.27	2.14	0.91
	Training time (s)	0.13	0.16	0.39	0.37	0.30	0.40	0.10	0.24	0.90	0.15	0.58
	Test time (s)	0.50	0.10	5.87	5.39	4.79	10.05	2.52	18.18	23.95	3.33	4.27
Lake	Parameters (M)	0.05	0.09	0.37	0.13	0.10	0.23	0.43	0.16	0.73	0.41	0.11
	FLOPs (G)	0.11	0.01	4.01	0.59	0.24	2.95	1.19	1.01	14.27	2.14	0.91
	Training time (s)	0.16	0.16	0.61	0.57	0.53	0.56	0.23	0.40	1.03	0.26	0.62
	Test time (s)	0.99	0.20	7.92	7.23	10.91	16.58	4.16	30.20	35.94	5.67	6.44
Bridge	Parameters (M)	0.05	0.09	0.37	0.13	0.10	0.23	0.43	0.16	0.73	0.41	0.11
	FLOPs (G)	0.11	0.01	4.01	0.59	0.24	2.95	1.19	1.02	14.27	2.14	0.92
	Training time (s)	0.16	0.15	0.50	0.51	0.45	0.57	0.21	0.36	1.07	0.24	0.60
	Test time (s)	0.91	0.10	6.23	5.61	7.87	14.92	3.74	27.21	35.95	4.76	6.37

adoption of both LS-CNN and CSS-CNN leads to performance improvements across all three datasets, with CSS-CNN in particular achieving gains in OA accuracy of 1.8%, 1.9%, and 1.1%, respectively, over the pure 3-D CNN approach. This further substantiates the efficacy of CSS-CNN for joint spatial-spectral feature extraction.

Regarding pooling strategies, the proposed DRCA module is evaluated against methods utilizing only max pooling or average pooling, underscoring its advantage in adaptive feature aggregation. Experimental results demonstrate that the DRCA module more effectively enhances the channelwise representation of spectral features and temperature features, achieving performance improvements of 0.5%, 1%, and 0.3%, respectively, compared to using max pooling or average pooling alone. Importantly, the AOFF module proves critical to OCSA-FN by facilitating deep feature mining and promoting cross-modal interactive weighting. Experimental results indicate that incorporating just the AOFF module resulted in OA improvements of 6.9%, 3.9%, and 4.4% across all the three datasets, respectively, highlighting its significant enhancement to model performance.

F. Complexity Analysis

For a quantitative analysis of the computational costs across various models, Table VIII provides an overview of the training time, test time, number of model parameters, and floating-point operations (FLOPs) for all three datasets.

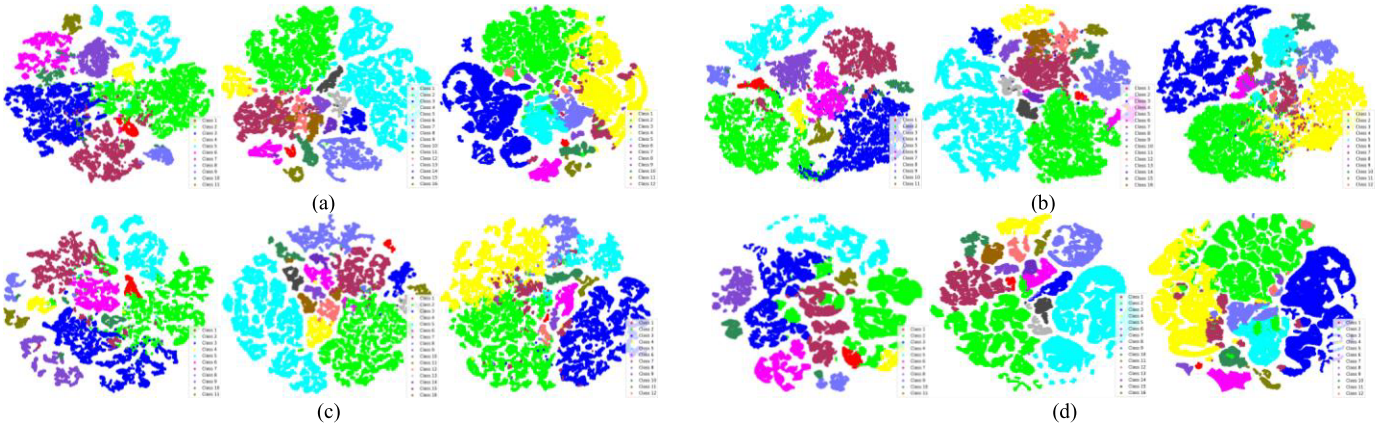


Fig. 13. T-SNE visualizations of feature representations learned by different methods on the three datasets. (a) CoupledCNN. (b) GLT-Net. (c) MASSFormer. (d) OCSA-FN. (Left) Urban dataset, (middle) lake dataset, and (right) bridge dataset.

The results demonstrate that while the proposed OCSA-FN exhibits moderate training time requirements, it significantly outperforms existing transformer-based methods as well as certain CNN-based fusion approaches in terms of parameter efficiency and computational complexity. This enhanced efficiency primarily stems from OCSA-FN's optimized feature interaction design, which substantially reduces computational overhead by eliminating redundant operations. For instance, compared to the MACN model, which has the fewest parameters and FLOPs, OCSA-FN demonstrates a 29.6% reduction in parameter count and a 9.4% decrease in FLOPs.

Such an efficient design not only enhances the model's computational effectiveness but also improves its practicality for practical applications.

G. Feature Visualization Analysis

To visually characterize the feature distributions learned by the model, this study employs t-distributed stochastic neighbor embedding (t-SNE) as a means to analyze and compare the feature representations extracted through various methods across three distinct datasets [58]. As illustrated in Fig. 13, we present t-SNE visualizations for different approaches, with each method evaluated on the Urban, Lake, and Bridge datasets (from left to right). The comparative results reveal that OCSA-FN effectively transforms spectral and temperature features derived from HSI and LST data into compact and highly discriminative representations, thereby providing a robust foundation for subsequent classification tasks. Additionally, the pronounced interclass separation and intraclass cohesion evident in the visualizations further underscore the efficacy of the features learned by our proposed method.

IV. Conclusion

This study proposes a convolutional spatial orthogonal attention network named OCSA-FN for HSI and LST data land cover classification, which effectively achieves deep interaction and comprehensive fusion of spectral and temperature features. The OCSA-FN model initially incorporates a DB-CNN module, in which the HSI branch extracts spectral features through a CSS-CNN sub-module, while the LST branch adaptively extracts temperature features using a LS-CNN sub-module. Subsequently, the DRCA module performs channelwise weighting of both spectral and temperature features via residual connections, which effectively alleviates gradient vanishing issues during the deep network training. The core innovation lies in the proposed AOFF module, which facilitates profound cross-modal interaction by uniquely integrating spatial orthogonal constraints with spatial attention mechanisms. This design successfully excavates complementary information between HSI and LST for enhanced classification performance.

This study conducts experiments on three real-world datasets. Through a series of parameter analysis experiments, the optimal network parameters are determined, including the patch size, learning rate, and the weighting coefficient λ associated with spatial orthogonality loss. Furthermore, comparative experiments are carried out against state-of-the-art transformer-based and CNN-based fusion classification methods. The results demonstrate that OCSA-FN surpasses the existing approaches in classification accuracy. In the ablation study, each proposed module surpasses to validate its individual contribution to overall performance. The findings indicate that AOFF module plays a pivotal role in improving classification performance by effectively exploring the commonalities and specificities within multisource data, thereby enhancing feature representation capability. Additionally, complexity analysis reveals that OCSA-FN achieves superior performance while utilizing fewer parameters and exhibiting lower FLOPs compared to existing transformer-based models, significantly reducing computational complexity. Improvements are also observed in both training and test time.

References

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] C. A. Bishop, J. G. Liu, and P. J. Mason, "Hyperspectral remote sensing for mineral exploration in Pulang, Yunnan Province, China," *Int. J. Remote Sens.*, vol. 32, no. 9, pp. 2409–2426, May 2011.

[3] E. L. Hestir et al., "Measuring freshwater aquatic ecosystems: The need for a hyperspectral global mapping satellite mission," *Remote Sens. Environ.*, vol. 167, pp. 181–195, Sep. 2015.

[4] J. Yuan, S. Wang, C. Wu, and Y. Xu, "Fine-grained classification of urban functional zones and landscape pattern analysis using hyperspectral satellite imagery: A case study of Wuhan," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3972–3991, 2022.

[5] C. Riefolo, A. Belmonte, R. Quarto, F. Quarto, S. Ruggieri, and A. Castrignano, "Potential of GPR data fusion with hyperspectral data for precision agriculture of the future," *Comput. Electron. Agricult.*, vol. 199, Aug. 2022, Art. no. 107109.

[6] Y. Wang, H. Wang, E. Zhao, M. Song, and C. Zhao, "Tucker decomposition-based network compression for anomaly detection with large-scale hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 10674–10689, 2024.

[7] Y. Wang, X. Chen, F. Wang, M. Song, and C. Yu, "Meta-learning based hyperspectral target detection using Siamese network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527913.

[8] X. Cao, H. Lin, S. Guo, T. Xiong, and L. Jiao, "Transformer-based masked autoencoder with contrastive loss for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524312.

[9] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[10] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[11] P. Ghamisi, J. A. Benediktsson, and S. Phinn, "Fusion of hyperspectral and LiDAR data in classification of urban areas," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 181–184.

[12] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[13] Q. Man, P. Dong, and H. Guo, "Pixel{-} and feature-level fusion of hyperspectral and LiDAR data for urban land-use classification," *Int. J. Remote Sens.*, vol. 36, no. 6, pp. 1618–1644, Mar. 2015.

[14] S. Jia et al., "Multiple feature-based superpixel-level decision fusion for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1437–1452, Feb. 2021.

[15] M. Zhang, P. Ghamisi, and W. Li, "Classification of hyperspectral and LiDAR data using extinction profiles with feature fusion," *Remote Sens. Lett.*, vol. 8, no. 10, pp. 957–966, Jun. 2017.

[16] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[17] H. R. Kalluri, S. Prasad, and L. M. Bruce, "Decision-level fusion of spectral reflectance and derivative information for robust hyperspectral land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4047–4058, Nov. 2010.

[18] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[19] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

[20] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.

[21] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal^{2}RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500114.

[22] Y. Wang, X. Chen, E. Zhao, and M. Song, "Self-supervised spectral-level contrastive learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510515.

[23] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[24] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[25] S. Fang, K. Li, and Z. Li, "S^{2}ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6504205.

[26] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[27] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5500205.

[28] M. E. Paoletti, J. M. Haut, S. K. Roy, and E. M. T. Hendrix, "Rotation equivariant convolutional neural networks for hyperspectral image classification," *IEEE Access*, vol. 8, pp. 179575–179591, 2020.

[29] M. Zhang, F. Gao, T. Zhang, Y. Gan, J. Dong, and H. Yu, "Attention fusion of transformer-based and scale-based method for hyperspectral and LiDAR joint classification," *Remote Sens.*, vol. 15, no. 3, p. 650, Jan. 2023.

[30] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514.

[31] E. Zhao, Y. Su, N. Qu, Y. Wang, C. Gao, and J. Zeng, "Self{-} and cross-attention enhanced transformer for visible and thermal infrared hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13408–13422, 2025.

[32] Y. Wang, X. Chen, E. Zhao, C. Zhao, M. Song, and C. Yu, "An unsupervised momentum contrastive learning based transformer network for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9053–9068, 2024.

[33] E. Zhao, N. Qu, Y. Wang, and C. Gao, "Spectral reconstruction from thermal infrared multispectral image using convolutional neural network and transformer joint network," *Remote Sens.*, vol. 16, no. 7, p. 1284, Apr. 2024.

[34] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.

[35] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[36] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[37] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[38] Y. He et al., "Multilevel attention dynamic-scale network for HSI and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5529916.

[39] X. Tang, Y. Zou, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Multiple information collaborative fusion network for joint classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5525416.

[40] M. Wang, Y. Sun, J. Xiang, and Y. Zhong, "CITNet: Convolution interaction transformer network for hyperspectral and LiDAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535918.

[41] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115.

[42] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415.

[43] X. Wang, J. Zhu, Y. Feng, and L. Wang, "MS2CANet: Multiscale spatial–spectral cross-modal attention network for hyperspectral image and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, 2024, Art. no. 5501505.

[44] Z. Wen and W. Yin, "A feasible method for optimization with orthogonality constraints," *Math. Program.*, vol. 142, nos. 1–2, pp. 397–434, Dec. 2013.

[45] R. Lai and S. Osher, "A splitting method for orthogonality constrained problems," *J. Sci. Comput.*, vol. 58, no. 2, pp. 431–449, Jun. 2013.

[46] E. Zhao et al., "An operational Land Surface Temperature retrieval methodology for Chinese second-generation huanjing disaster monitoring satellite data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1283–1292, 2022.

[47] E. Zhao, N. Qu, Y. Wang, C. Gao, and J. Zeng, "TEBS: Temperature–emissivity–driven band selection for thermal infrared hyperspectral image classification with structured state-space model and gated attention," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 142, Oct. 2025, Art. no. 104710.
[48] E. Zhao et al., "Thermal infrared hyperspectral band selection via graph neural network for Land Surface Temperature retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5003414.
[49] Z. Qin, A. Karnieli, and P. Berliner, "A mono-window algorithm for retrieving Land Surface Temperature from Landsat TM data and its application to the Israel–Egypt border region," *Int. J. Remote Sens.*, vol. 22, no. 18, pp. 3719–3746, Nov. 2010.
[50] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2012.
[51] A. Gillespie, S. Rokugawa, T. Matsunaga, J. S. Cothern, S. Hook, and A. B. Kahle, "A temperature and emissivity separation algorithm for Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) images," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 4, pp. 1113–1126, Jul. 1998.
[52] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.
[53] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.
[54] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.
[55] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.
[56] K. Li, D. Wang, X. Wang, G. Liu, Z. Wu, and Q. Wang, "Mixing self-attention and convolution: A unified framework for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5523216.
[57] L. Sun, H. Zhang, Y. Zheng, Z. Wu, Z. Ye, and H. Zhao, "MASSFormer: Memory-augmented spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516415.
[58] L. V. D. Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Nov. 2008.



Enyu Zhao (Member, IEEE) was born in Dalian, Liaoning, China, in 1987. He received the Ph.D. degree in cartography and geographic information system from the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing, China, in 2017.

He was a Joint Ph.D. Student with the Engineering Science, Computer Science and Imaging Laboratory, University of Strasbourg, Strasbourg, France, from 2014 to 2016. He is currently an Associate Professor with the College of Information Science and Technology, Dalian Maritime University, Dalian. His research interests include quantitative remote sensing and hyperspectral image processing.



Yongfang Su was born in Shangqiu, Henan, China, in 2000. He received the B.S. degree in automation from Xi'an University of Technology, Xi'an, China, in 2023. He is currently pursuing the M.S. degree in computer science and technology with Dalian Maritime University, Dalian, Liaoning, China.

His research interests include hyperspectral image classification and deep learning.



Nianxin Qu (Student Member, IEEE) was born in Liaoyang, Liaoning, China, in 1999. He received the M.S. degree in computer science and technology from Dalian Maritime University, Dalian, Liaoning, China, in 2024, where he is currently pursuing the Ph.D. degree in computer science and technology.

His research interests include hyperspectral image processing and deep learning.



Yulei Wang (Member, IEEE) was born in Yantai, Shandong, China, in 1986. She received the B.S. and Ph.D. degrees in signal and information processing from Harbin Engineering University, Harbin, China, in 2009 and 2015, respectively.

She was a Joint Ph.D. Student at the Remote Sensing Signal and Image Processing Laboratory, University of Maryland, Baltimore County, Baltimore, MD, USA, from 2011 to 2013. From 2011 to 2013, she was a Research Assistant with the Shock, Trauma, and Anesthesiology Research Organized Research Center (STAR-ORC), School of Medicine, University of Maryland, College Park, MD, USA. She is currently an Associate Professor and a Doctoral Supervisor at the Hyperspectral Imaging in Remote Sensing (CHIRS), Information Science and Technology College, Dalian Maritime University, Dalian, China. Her research interests include hyperspectral image processing, multisource remote sensing fusion and vital signs signal processing. More details could be found at https://YuleiWang1.github.io/



Yongguang Zhao received the B.S. degree in geographic information system from Central South University, Changsha, China, in 2009, and the Ph.D. degree in signal and information processing from the Academy of Opto-Electronics, Chinese Academy of Sciences (CAS), Beijing, China, in 2015.

He is currently an Associate Professor with the Aerospace Information Research Institute, CAS. His research interests include radiometric calibration and quantitative remote sensing application.